Bug Reports — HW7: IoT Network / Memory Stability

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Defects found while testing network loss/recovery, the offline buffer and memory stability of the IoT node. The node recovers from network loss by itself, but four of eight requirements fail (R2, R4, R6, R7).

Environment: Wokwi Simulator in VS Code + PlatformIO; firmware: homework_buggy; OS: Windows 11

Summary

ID	Title	Severity
BUG-01	Reconnect backoff does not grow: after NET DOWN the node retries every ~50 ms (reconnect storm)	Major
BUG-02	Every buffered message is published twice after NET UP (duplicate publish during flush)	Major
BUG-03	Buffer overflow: capacity exceeded, DROP_OLDEST not applied, oldest messages lost without dropped counter, flush sends a corrupted set	Critical
BUG-04	Memory leak: free heap drops by ≈80 bytes per message and never recovers	Critical

BUG-01 — Reconnect backoff does not grow: after NET DOWN the node retries every ~50 ms (reconnect storm)

Requirement / Risk	R2 / RISK-02
Environment	Wokwi Simulator in VS Code + PlatformIO; firmware: homework_buggy; OS: Windows 11
Preconditions	Node CONNECTED, STATS reset (RESET STATS), STATUS shows reconnect_attempts=0.
Steps	1. RESET STATS 2. NET DOWN 3. Wait 10–15 s and watch the [NET] retry=… backoff_ms=… lines 4. STATUS 5. NET UP
Expected	The pause between reconnect attempts increases (e.g. 500 → 1000 → 2000 ms, capped) and does not create a reconnect storm (R2).
Actual	The first pause is 500 ms (retry=1); from retry=2 on, backoff_ms is always 50 ms. 204 attempts in ≈13.7 s (≈20 attempts per second). The log flood makes the serial console hard to use.
Recovery behavior	The node recovers by itself: after NET UP it reaches CONNECTED in ≈2 ms, reconnect=SUCCESS, NET LED on.
IDs / Counters / Timing	retry=1 at 13:19:51.818 (backoff_ms=500); retry=204 at 13:20:05.543 (backoff_ms=50). retry 2–125: 123 attempts in 6.179 s = 50.2 ms avg; retry 126–204: 78 attempts in 3.933 s = 50.4 ms avg. STATUS during outage: {"net":"BACKOFF","broker":false,…,"reconnect_attempts":125,"heap":350868}
Heap before/after	350868 / 350868 (no change)

Evidence	`[NET] retry=1 backoff_ms=500`
	`[NET] retry=2 backoff_ms=50`
	`[NET] retry=3 backoff_ms=50 … [NET] retry=204 backoff_ms=50`

	`Full log: platformio-device-monitor-*.log, 13:19:51–13:20:08`
Reproducibility	2/2 detailed runs with timestamps; the same 50 ms loop was seen in every later offline test (e.g. 639 attempts in ≈80 s).
Severity rationale	Major. On real hardware a 20 Hz reconnect loop loads the broker and Wi-Fi and drains the battery; there is no workaround on the device side.
Severity	Major

BUG-02 — Every buffered message is published twice after NET UP (duplicate publish during flush)

Requirement / Risk	R6 / RISK-05
Environment	Wokwi Simulator in VS Code + PlatformIO; firmware: homework_buggy; OS: Windows 11
Preconditions	Node CONNECTED, buffer=0, after RESET STATS.
Steps	1. RESET STATS 2. NET DOWN 3. SEND (5 times, one command each) 4. BUFFER (count=5) 5. NET UP 6. STATUS
Expected	Each message ID is successfully published once: sent=5, duplicates=0 (R6).
Actual	Every buffered ID appears twice in the [MQTT] publish lines. STATUS: sent=10, duplicates=5. Not reproducible while CONNECTED (STRESS 100 ×10: sent=1000, duplicates=0), so the defect is in the flush after reconnect.
Recovery behavior	The node recovers by itself and the buffer becomes 0, but duplicated data is sent.
IDs / Counters / Timing	IDs 4–8: each published 2×, 13:40:24.719–13:40:24.730. flush_start count=5 → flush_end count=0. Other runs: 3 messages → sent=6, duplicates=3; 8 messages → sent=16, duplicates=8; 9 messages → sent=18, duplicates=9.
Heap before/after	350364 → 349964 (after the 5-message flush; see BUG-04)

Evidence	`[BUFFER] flush_start count=5`
	`[MQTT] publish id=4 temp=24.5`
	`[MQTT] publish id=4 temp=24.5`
	`[MQTT] publish id=5 temp=24.5`
	`[MQTT] publish id=5 temp=24.5 …`
	`{"net":"CONNECTED","broker":true,"buffer":0,"capacity":8,"sent":10,"dropped":0,"duplicates":5,"reconnect_attempts":333,"heap":349964}`

Reproducibility	5/5 (every flush after reconnect in runs B ×2 and C ×3)
Severity rationale	Major. Telemetry is duplicated on the broker, which breaks counting and analytics; no workaround on the device.
Severity	Major

BUG-03 — Buffer overflow: capacity exceeded, DROP_OLDEST not applied, oldest messages lost without dropped counter, flush sends a corrupted set

Requirement / Risk	R4 (and R6, R8) / RISK-04
Environment	Wokwi Simulator in VS Code + PlatformIO; firmware: homework_buggy; OS: Windows 11
Preconditions	Node CONNECTED, buffer=0, after RESET STATS; buffer capacity = 8.
Steps	1. RESET STATS 2. NET DOWN 3. SEND ×9 (also repeated with ×12) 4. BUFFER, STATUS

	5. NET UP 6. STATUS
Expected	The buffer never exceeds 8; with 9 messages dropped=1 (with 12: dropped=4) and a log entry; after NET UP the 8 newest IDs are published once each; no corrupted data (R4).
Actual	9 messages: queued id=37 count=9/8, BUFFER count=9 capacity=8 dropped=0; the log says "overflow policy=DROP_OLDEST dropped_total=0" but nothing is dropped. The flush sends 37, 30, 31 … 36, 37 (ID 29 never published, ID 37 twice in the buffer), every publish duplicated; STATUS sent=18, dropped=0. 12 messages: count=12/8, dropped=0; the flush publishes 17–20 (four times each), 13–16 (twice), IDs 9–12 never published. 8 messages: works (count=8/8, dropped=0).
Recovery behavior	The node reconnects by itself, but the lost messages are never reported.
IDs / Counters / Timing	N=9: IDs 29–37; 14:07:38 overflow log; 14:07:46 BUFFER count=9 dropped=0; flush 14:07:51.650–.672; STATUS sent=18 dropped=0 duplicates=9. N=12: IDs 9–20; 13:51:05.889 overflow log; 13:51:28 BUFFER count=12 dropped=0; STATUS (offline) buffer=12 capacity=8; flush 13:51:45.166–.221 (flush_start count=12).
Heap before/after	N=9: 348364 → 347644; N=12: 349004 before flush

Evidence	`[BUFFER] queued id=36 count=8/8`
	`[BUFFER] overflow policy=DROP_OLDEST dropped_total=0`
	`[BUFFER] queued id=37 count=9/8`
	`[BUFFER] count=9 capacity=8 dropped=0`
	`[BUFFER] flush_start count=9`
	`[MQTT] publish id=37 … id=30 … id=31 … id=36 … id=37`
	`{"buffer":0,"capacity":8,"sent":18,"dropped":0,"duplicates":9,…}`

Reproducibility	2/2 (N=9 and N=12); the boundary N=8 passes
Severity rationale	Critical. Data is lost silently (the dropped counter stays 0), the buffer grows beyond its capacity, and the flushed data is wrong.
Severity	Critical

BUG-04 — Memory leak: free heap drops by ≈80 bytes per message and never recovers

Requirement / Risk	R7 / RISK-06
Environment	Wokwi Simulator in VS Code + PlatformIO; firmware: homework_buggy; OS: Windows 11
Preconditions	Simulator freshly restarted; node CONNECTED; RESET STATS; STATUS gives baseline heap=350868.
Steps	1. RESET STATS, STATUS (iteration 0) 2. STRESS 100, STATUS 3. Repeat step 2 ten times in total
Expected	No systematic irreversible drop of free heap on repeated SEND/STRESS; small fluctuations in both directions are acceptable (R7).
Actual	Heap decreases on every iteration by ≈8000 bytes (≈80 bytes per message) and never goes back: 350868 → 270600 after 1000 messages (−80268 bytes, ≈23%). At this rate memory would be exhausted after ≈4400 messages (extrapolation).
Recovery behavior	Heap is not released after the stress finishes; it returns to the baseline only after a simulator restart.
IDs / Counters / Timing	Δheap per iteration: −8264, −8000, −7992, −8012, −8000, −8000, −8000, −8000, −8000, −8000. Each iteration: sent += 100, dropped=0, duplicates=0, reconnect_attempts=0.
Heap before/after	350868 → 270600

Evidence	`[STRESS] start count=100 heap=350868`
	`[STRESS] end heap=342604`
	`[STRESS] start count=100 heap=342604`
	`[STRESS] end heap=334604 … [STRESS] end heap=270600`

	Log 14:13:43–14:16:20
Reproducibility	10/10 iterations in one run; heap also fell after every flush in scenarios B and C (350868 → 347644).
Severity rationale	Critical. The node stops working after enough messages (memory exhaustion).
Severity	Critical